Experiment 2

Exploratory Data Analysis, Descriptive Statistics, and Data Visualization

Subject: Data Science Lab | Dataset: Titanic | Mapped CO(s): CO1, CO2

1. Aim

To perform EDA, descriptive statistical analysis, and data visualization on the Titanic dataset.

2. Key Concepts & Definitions

Exploratory Data Analysis (EDA)

Examining data to understand structure, patterns, and missing values before modeling.

Mean (x-bar)

Average value - sum of all observations divided by count.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Median

Middle value when sorted. Robust to outliers.

Mode

Most frequently occurring value.

Variance (s^2)

Average squared deviation from the mean.

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

Standard Deviation (sigma)

Square root of variance.

$$\sigma = \sqrt{s^2}$$

Quartiles

Q1 = 25th percentile, Q2 = median (50th), Q3 = 75th percentile.

Interquartile Range (IQR)

Spread of the middle 50% of data.

$$\text{IQR} = Q_3 - Q_1$$

Outlier

Extreme value far from the rest of the data.

$$x < Q_1 - 1.5\,\text{IQR} \quad \text{or} \quad x > Q_3 + 1.5\,\text{IQR}$$

Skewness

Asymmetry of distribution. Right-skewed: mean > median.

Missing Value

Empty/NaN entry. Must be identified before analysis.

$$\text{Missing \%} = \frac{\text{Missing Count}}{\text{Total Rows}} \times 100$$

Covariance

How two variables change together.

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

Correlation (r)

Standardized linear relationship measure.

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

3. Symbol Glossary

Symbol	Meaning
n	Number of observations
x_i	i-th data value
x-bar	Sample mean
s^2	Sample variance
sigma	Standard deviation
Q1, Q2, Q3	First, second (median), third quartile
IQR	Interquartile range
Cov(X,Y)	Covariance
r	Pearson correlation

4. Important Formulae (Summary)

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$\sigma = \sqrt{s^2}$$

$$\text{IQR} = Q_3 - Q_1$$

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper fence} = Q_3 + 1.5 \times \text{IQR}$$

$$\text{Missing \%} = \frac{\text{Missing Count}}{\text{Total Rows}} \times 100$$

$$\text{Cov}(X, Y) = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

5. Algorithm

- 1. Import libraries; load Titanic dataset
- 2. Display head(), info(), dtypes; check missing values

- 3. Generate describe(); plot bar charts, histograms, box plots
- 4. Create scatter plot and correlation heatmap; interpret results

6. Experimental Results

- Shape: (891, 15)

Missing Values

Column	Missing	Missing %
deck	688	77.22
age	177	19.87
embarked	2	0.22

Descriptive Statistics

Variable	Mean	Std Dev	Min	Max
age	29.70	14.53	0.42	80.00
fare	32.20	49.69	0.00	512.33
sibsp	0.52	1.10	0	8

Outlier Analysis - Fare

- IQR = 23.09, Upper fence = 65.63, Outliers = 116

7. Visualizations

- Bar Chart: Categorical distributions (sex, class, survived)
- Histogram: Numerical distributions (Age, Fare)
- Box Plot: Median, quartiles, outliers
- Heatmap: Correlation matrix of numerical variables

8. Viva Q&A

Q: What is variance?

A: Var(X) = (1/(n-1)) sum of (x_i - x-bar)^2 - measures spread.

Q: What is IQR?

A: Q3 - Q1 - spread of the middle 50% of data.

Q: What is covariance?

A: Cov(X,Y) - measures co-movement of two variables (see formula).

Q: Mean vs median?

A: Mean is average; median is middle value. Median is robust to outliers.

9. Conclusion

EDA and visualization successfully revealed structure, missing values, outliers, and relationships in the Titanic dataset.